

Assignment 3 - Data visualization for economics

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First Part:

```
library(rvest)
library(tidyverse)
library(stringr)
```

Q1. Scrape the tax table from Wikipedia

```
# Wikipedia URL
url <- "https://en.wikipedia.org/wiki/List_of_countries_by_tax_rates"

# Read html table
page <- read_html(url)

# Extract all wikitable elements
tables <- page %>% html_elements("table.wikitable")

length(tables)
```

```
#> [1] 1
```

```
# Scrape 1st table:
tax_raw <- tables[[1]] %>%
  html_table(fill = TRUE)
head(tax_raw)
```

```
#> # A tibble: 6 x 10
#>   `Tax jurisdiction` Corporate `Individual income` `Individual income`
#>   <chr>                <chr>      <chr>                <chr>
#> 1 Tax jurisdiction    Corporate Lowest                Highest
#> 2 Afghanistan        20%[2]    0%[3]                20%[3]
#> 3 Albania             15%[6]    0%[7]                23%[7]
#> 4 Algeria             19 - 26%[9] 0%[10]                35%[10]
#> 5 American Samoa      34%[6]     4%[12] [13]          6%[13]
#> 6 Andorra             10%[15]    0%[16]                10%[16]
#> # i 6 more variables: `Capital gains[1]` <chr>, Wealth <chr>, Property <chr>,
#> #   `Inheritance / Estate` <chr>, `VAT or GSTor Sales` <chr>,
#> #   `Further reading` <chr>
```

```
# Remove the header row that is repeated inside the table
tax_raw <- tax_raw[-1, ]
head(tax_raw)
```

```
#> # A tibble: 6 x 10
#>   `Tax jurisdiction` Corporate `Individual income` `Individual income`
#>   <chr>                <chr>      <chr>                <chr>
```

```
#> 1 Afghanistan      20%[2]      0%[3]      20%[3]
#> 2 Albania           15%[6]      0%[7]      23%[7]
#> 3 Algeria           19 - 26%[9]  0%[10]     35%[10]
#> 4 American Samoa    34%[6]      4%[12][13]  6%[13]
#> 5 Andorra           10%[15]     0%[16]     10%[16]
#> 6 Angola            30%[18]     0%[18]     17%[18]
#> # i 6 more variables: `Capital gains[1]` <chr>, Wealth <chr>, Property <chr>,
#> # `Inheritance / Estate` <chr>, `VAT or GSTor Sales` <chr>,
#> # `Further reading` <chr>
```

Q2. Rename columns

```
names(tax_raw)
```

```
#> [1] "Tax jurisdiction"      "Corporate"            "Individual income"
#> [4] "Individual income"     "Capital gains[1]"     "Wealth"
#> [7] "Property"             "Inheritance / Estate" "VAT or GSTor Sales"
#> [10] "Further reading"
```

```
# Step 1: fix duplicate column names
```

```
names(tax_raw)[names(tax_raw) == "Individual income"] <-
  c("Individual income lowest", "Individual income highest")
```

```
# Step 2: rename to beautiful human-readable names
```

```
tax_renamed <- tax_raw %>%
  rename(
    `Country` = `Tax jurisdiction`,
    `Corporate Tax Rate` = Corporate,
    `Lowest Income Tax Rate` = `Individual income lowest`,
    `Top Marginal Income Tax Rate` = `Individual income highest`,
    `Capital Gains Tax Rate` = `Capital gains[1]`,
    `Wealth Tax` = Wealth,
    `Property Tax` = Property,
    `Inheritance or Estate Tax` = `Inheritance / Estate`,
    `VAT Rate` = `VAT or GSTor Sales`,
    `Further Reading` = `Further reading`
  )
```

```
colnames(tax_renamed)
```

```
#> [1] "Country"                "Corporate Tax Rate"
#> [3] "Lowest Income Tax Rate" "Top Marginal Income Tax Rate"
#> [5] "Capital Gains Tax Rate" "Wealth Tax"
#> [7] "Property Tax"          "Inheritance or Estate Tax"
#> [9] "VAT Rate"              "Further Reading"
```

Q3. removing footnotes and non-numeric characters

```
library(stringr)
```

```
library(purrr)
```

```
clean_number <- function(x) {
  x %>%
    as.character() %>%
    str_remove_all("\\[[^]]*\\]") %>% # remove footnote[9]
    str_replace_all("-", "-") %>% # — => -
    str_extract_all("\\d+\\.?\\d*") %>% # extract number
    map_dbl(function(nums) {
      nums <- as.numeric(nums)
    })
}
```

```

# only keep <100
nums <- nums[nums <= 100]

if (length(nums) == 0) {
  return(NA_real_)
} else if (length(nums) == 1) {
  return(nums[1])
} else {
  return(mean(nums)) # intervalle => mean
}
})
}
clean_number

```

```

#> function (x)
#> {
#>   x %>% as.character() %>% str_remove_all("\\[[^]]*\\]") %>%
#>     str_replace_all("-", "-") %>% str_extract_all("\\d+\\.?\\d*") %>%
#>     map_dbl(function(nums) {
#>       nums <- as.numeric(nums)
#>       nums <- nums[nums <= 100]
#>       if (length(nums) == 0) {
#>         return(NA_real_)
#>       }
#>       else if (length(nums) == 1) {
#>         return(nums[1])
#>       }
#>       else {
#>         return(mean(nums))
#>       }
#>     })
#> }

```

```

# Apply cleaning function to all tax rate columns
tax_clean <- tax_renamed %>%
  mutate(
    `Corporate Tax Rate` = clean_number(`Corporate Tax Rate`),
    `Lowest Income Tax Rate` = clean_number(`Lowest Income Tax Rate`),
    `Top Marginal Income Tax Rate` = clean_number(`Top Marginal Income Tax Rate`),
    `Capital Gains Tax Rate` = clean_number(`Capital Gains Tax Rate`),
    `Wealth Tax` = clean_number(`Wealth Tax`),
    `Property Tax` = clean_number(`Property Tax`),
    `Inheritance or Estate Tax` = clean_number(`Inheritance or Estate Tax`),
    `VAT Rate` = clean_number(`VAT Rate`)
  )
tax_clean

```

```

#> # A tibble: 232 x 10
#>   Country      `Corporate Tax Rate` Lowest Income Tax Ra~1 Top Marginal Income ~2
#>   <chr>          <dbl>          <dbl>          <dbl>
#> 1 Afghanistan~      20              0              20
#> 2 Albania        15              0              23
#> 3 Algeria        22.5             0              35
#> 4 American ~     34              4              6
#> 5 Andorra         10              0             10
#> 6 Angola         30              0             17
#> 7 Anguilla        0              0              0
#> 8 Antigua a~     25              0              0
#> 9 Argentina       25              9             35
#> 10 Armenia        18             22             22
#> # i 222 more rows
#> # i abbreviated names: 1: `Lowest Income Tax Rate`,
#> #   2: `Top Marginal Income Tax Rate`

```

```
#> # i 6 more variables: `Capital Gains Tax Rate` <dbl>, `Wealth Tax` <dbl>,
#> # `Property Tax` <dbl>, `Inheritance or Estate Tax` <dbl>, `VAT Rate` <dbl>,
#> # `Further Reading` <chr>
```

```
# Check tax variables are now numeric
str(tax_clean)
```

```
#> tibble [232 x 10] (S3: tbl_df/tbl/data.frame)
#> $ Country                : chr [1:232] "Afghanistan" "Albania" "Algeria" "American Samoa" ...
#> $ Corporate Tax Rate      : num [1:232] 20 15 22.5 34 10 30 0 25 25 18 ...
#> $ Lowest Income Tax Rate  : num [1:232] 0 0 0 4 0 0 0 0 9 22 ...
#> $ Top Marginal Income Tax Rate: num [1:232] 20 23 35 6 10 17 0 0 35 22 ...
#> $ Capital Gains Tax Rate   : num [1:232] NA 15 17.5 NA NA 10 NA NA 15 15 ...
#> $ Wealth Tax              : num [1:232] NA NA NA NA NA NA NA NA NA NA ...
#> $ Property Tax            : num [1:232] NA NA NA 0 NA NA NA NA NA NA ...
#> $ Inheritance or Estate Tax : num [1:232] NA NA NA NA NA NA NA NA NA NA ...
#> $ VAT Rate                : num [1:232] 0 13 14 0 3.5 14 0 15 21 20 ...
#> $ Further Reading         : chr [1:232] "Taxation in Afghanistan" "Taxation in Albania" "Taxation
```

```
summarise(tax_clean)
```

```
#> # A tibble: 1 x 0
```

Q4. Add a Continent column

```
library(countrycode)

tax_with_continent <- tax_clean %>%
  mutate(
    Country = Country %>%
      str_remove_all("\\[[~]]*\\") %>% #remove footnote
      str_trim(), #remove whitespace

    # harmonise a few special cases so that countrycode can recognise them
    Country = case_when(
      Country == "United States" ~ "USA",
      Country == "United Kingdom" ~ "UK",
      Country == "People's Republic of China" ~ "China",
      Country == "Republic of Korea" ~ "South Korea",
      Country == "Azores" ~ "Portugal",
      Country == "Madeira" ~ "Portugal",
      Country == "Micronesia" ~ "Federated States of Micronesia",
      Country == "Saint Martin" ~ "Saint Martin (French part)",
      Country == "Sark" ~ "United Kingdom",
      TRUE ~ Country
    ),

    # add continent information
    Continent = countrycode(
      Country,
      origin = "country.name",
      destination = "continent",
      custom_match = c("Kosovo" = "Europe")
    )
  )

tax_with_continent
```

```
#> # A tibble: 232 x 11
#>   Country `Corporate Tax Rate` Lowest Income Tax Ra~1 Top Marginal Income ~2
#>   <chr>                <dbl>                <dbl>                <dbl>
```

```
#> 1 Afghanist~      20      0      20
#> 2 Albania        15      0      23
#> 3 Algeria        22.5    0      35
#> 4 American ~     34      4       6
#> 5 Andorra         10      0      10
#> 6 Angola          30      0      17
#> 7 Anguilla         0      0       0
#> 8 Antigua a~      25      0       0
#> 9 Argentina       25      9      35
#> 10 Armenia        18     22      22
#> # i 222 more rows
#> # i abbreviated names: 1: `Lowest Income Tax Rate`,
#> #   2: `Top Marginal Income Tax Rate`
#> # i 7 more variables: `Capital Gains Tax Rate` <dbl>, `Wealth Tax` <dbl>,
#> #   `Property Tax` <dbl>, `Inheritance or Estate Tax` <dbl>, `VAT Rate` <dbl>,
#> #   `Further Reading` <chr>, Continent <chr>
```

```
# Check no unmatched countries
tax_with_continent %>%
  filter(is.na(Continent)) %>%
  select(Country)
```

```
#> # A tibble: 0 x 1
#> # i 1 variable: Country <chr>
```

Q5. Average tax rates by continent

```
avg_by_continent <- tax_with_continent %>%
  group_by(Continent) %>%
  summarise(
    AvgTopIncome = mean(`Top Marginal Income Tax Rate`, na.rm = TRUE),
    AvgVAT = mean(`VAT Rate`, na.rm = TRUE),
    AvgCorporate = mean(`Corporate Tax Rate`, na.rm = TRUE)
  )

# order by average top marginal income tax rate (descending)
avg_by_continent %>%
  arrange(desc(AvgTopIncome))
```

```
#> # A tibble: 5 x 4
#>   Continent AvgTopIncome AvgVAT AvgCorporate
#>   <chr>      <dbl>    <dbl>    <dbl>
#> 1 Africa      32.2    13.8      27.7
#> 2 Europe      27.0    11.1      14.9
#> 3 Americas    24.9     9.81     22.5
#> 4 Asia        20.3     8.92     17.6
#> 5 Oceania     18.4     5.95     18.0
```

```
# order by average VAT rate (descending)
avg_by_continent %>%
  arrange(desc(AvgVAT))
```

```
#> # A tibble: 5 x 4
#>   Continent AvgTopIncome AvgVAT AvgCorporate
#>   <chr>      <dbl>    <dbl>    <dbl>
#> 1 Africa      32.2    13.8      27.7
#> 2 Europe      27.0    11.1      14.9
#> 3 Americas    24.9     9.81     22.5
#> 4 Asia        20.3     8.92     17.6
#> 5 Oceania     18.4     5.95     18.0
```

```
# order by corporate tax rate (descending)
avg_by_continent %>%
  arrange(desc(AvgCorporate))
```

```
#> # A tibble: 5 x 4
#>   Continent AvgTopIncome AvgVAT AvgCorporate
#>   <chr>      <dbl>   <dbl>      <dbl>
#> 1 Africa      32.2    13.8        27.7
#> 2 Americas    24.9     9.81       22.5
#> 3 Oceania     18.4     5.95       18.0
#> 4 Asia        20.3     8.92       17.6
#> 5 Europe      27.0    11.1        14.9
```

Africa has the highest average top marginal income tax rate (around 32.2%), followed by Europe (about 27.0%).

Africa also records the highest average VAT rate (approximately 13.8%), while Oceania has the lowest VAT level (around 6%).

For comparison, we additionally report the average corporate tax rate by continent: Africa again has the highest value (27.7%), whereas Europe has the lowest (14.9%).

Q6. Bar chart of average top marginal income tax rate by continent

```
library(ggplot2)

p6 <- avg_by_continent %>%
  ggplot(aes(x = Continent,
             y = AvgTopIncome,
             fill = Continent)) +

  geom_col(show.legend = FALSE) +

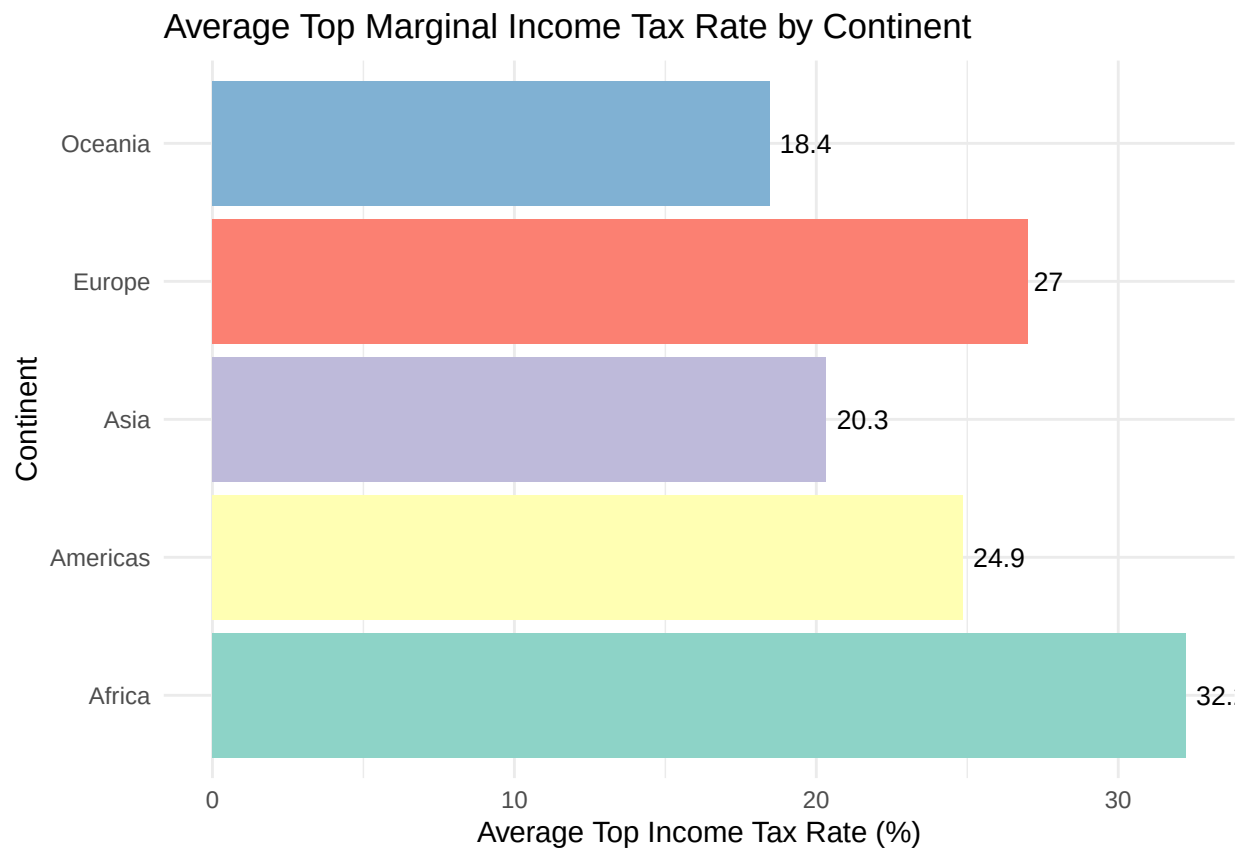
  coord_flip() +

  geom_text(
    aes(label = round(AvgTopIncome, 1)),
    hjust = -0.2,
    size = 3.5,
    color = "black"
  ) +

  scale_fill_brewer(palette = "Set3") +

  labs(
    title = "Average Top Marginal Income Tax Rate by Continent",
    x = "Continent",
    y = "Average Top Income Tax Rate (%)"
  ) +
  theme_minimal()
```

p6



Q7. Grouped bar chart: VAT vs Top Income Tax

```
library(tidyr)

plot_data <- avg_by_continent %>%
  pivot_longer(
    cols = c(AvgTopIncome, AvgVAT),
    names_to = "TaxType",
    values_to = "Rate"
  ) %>%
  mutate(
    TaxType = recode(
      TaxType,
      "AvgTopIncome" = "Top marginal income tax",
      "AvgVAT" = "VAT"
    )
  )
plot_data
```

```
#> # A tibble: 10 x 4
#>   Continent AvgCorporate TaxType      Rate
#>   <chr>      <dbl> <chr>      <dbl>
#> 1 Africa      27.7 Top marginal income tax 32.2
#> 2 Africa      27.7 VAT                13.8
#> 3 Americas    22.5 Top marginal income tax 24.9
#> 4 Americas    22.5 VAT                9.81
#> 5 Asia        17.6 Top marginal income tax 20.3
#> 6 Asia        17.6 VAT                8.92
#> 7 Europe      14.9 Top marginal income tax 27.0
#> 8 Europe      14.9 VAT                11.1
#> 9 Oceania     18.0 Top marginal income tax 18.4
#> 10 Oceania    18.0 VAT                5.95
```

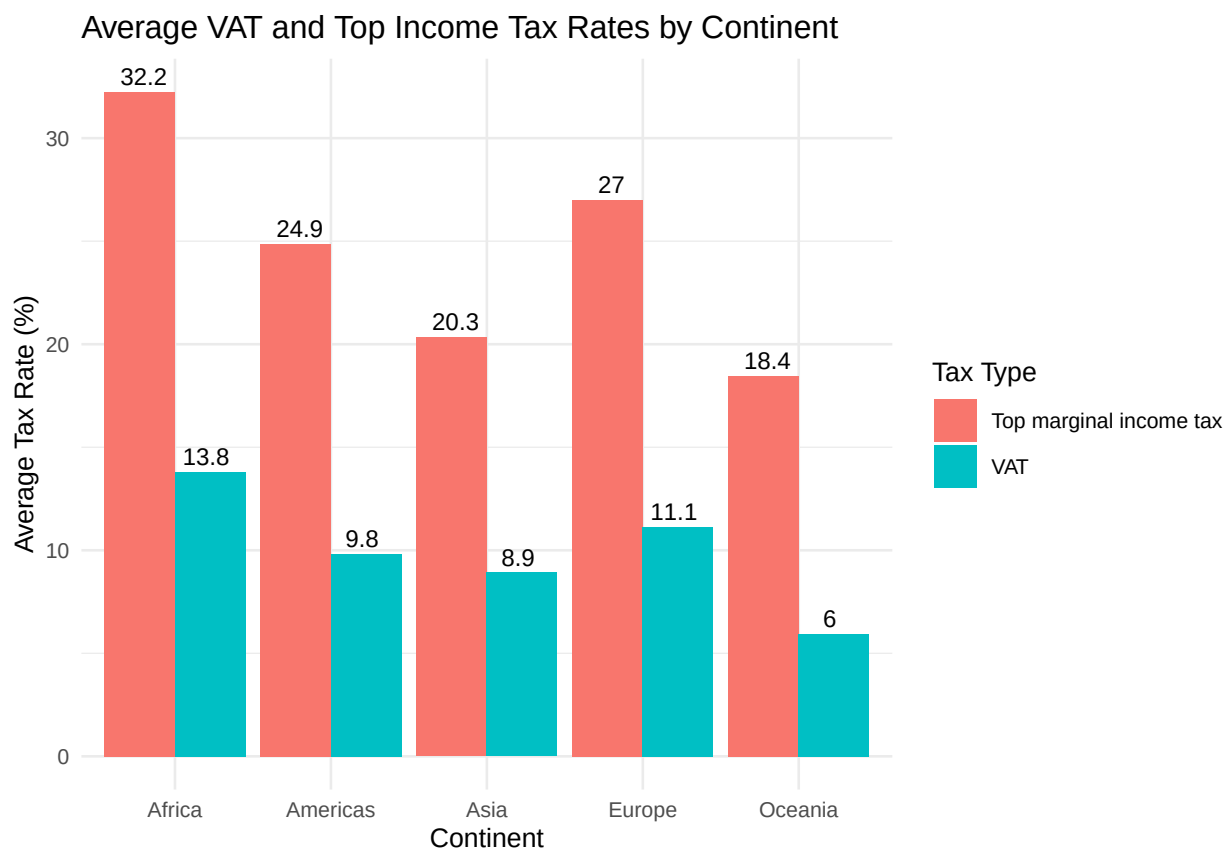
```

p7 <- ggplot(plot_data,
             aes(x = Continent,
                 y = Rate,
                 fill = TaxType)) +
  geom_col(position = "dodge") +

  geom_text(
    aes(label = round(Rate, 1)),
    position = position_dodge(width = 0.8),
    vjust = -0.4,
    size = 3
  ) +

  labs(
    title = "Average VAT and Top Income Tax Rates by Continent",
    x = "Continent",
    y = "Average Tax Rate (%)",
    fill = "Tax Type"
  )
p7

```



Q8. Map: Top Marginal Income Tax Rate by Country

```

library(maps)
library(tidyverse)

# step1: map data
world_map <- map_data("world")

# step2: combined tax and map data
map_data_join <- world_map %>%
  left_join(tax_with_continent,
            by = c("region" = "Country"))
head(map_data_join)

```



```

#>      long      lat group order region subregion Corporate Tax Rate
#> 1 -69.89912 12.45200     1     1  Aruba      <NA>                25
#> 2 -69.89571 12.42300     1     2  Aruba      <NA>                25
#> 3 -69.94219 12.43853     1     3  Aruba      <NA>                25
#> 4 -70.00415 12.50049     1     4  Aruba      <NA>                25
#> 5 -70.06612 12.54697     1     5  Aruba      <NA>                25
#> 6 -70.05088 12.59707     1     6  Aruba      <NA>                25
#> Lowest Income Tax Rate Top Marginal Income Tax Rate Capital Gains Tax Rate
#> 1                      7                      58.95                NA
#> 2                      7                      58.95                NA
#> 3                      7                      58.95                NA
#> 4                      7                      58.95                NA
#> 5                      7                      58.95                NA
#> 6                      7                      58.95                NA
#> Wealth Tax Property Tax Inheritance or Estate Tax VAT Rate Further Reading
#> 1      NA            NA                      NA      1.5 Taxation in Aruba
#> 2      NA            NA                      NA      1.5 Taxation in Aruba
#> 3      NA            NA                      NA      1.5 Taxation in Aruba
#> 4      NA            NA                      NA      1.5 Taxation in Aruba
#> 5      NA            NA                      NA      1.5 Taxation in Aruba
#> 6      NA            NA                      NA      1.5 Taxation in Aruba
#> Continent
#> 1 Americas
#> 2 Americas
#> 3 Americas
#> 4 Americas
#> 5 Americas
#> 6 Americas

```

```

# stpe3: plot
p8 <- ggplot(map_data_join,
  aes(x = long,
    y = lat,
    group = group,
    fill = `Top Marginal Income Tax Rate`)) +

  geom_polygon(color = "gray70", linewidth = 0.1) +

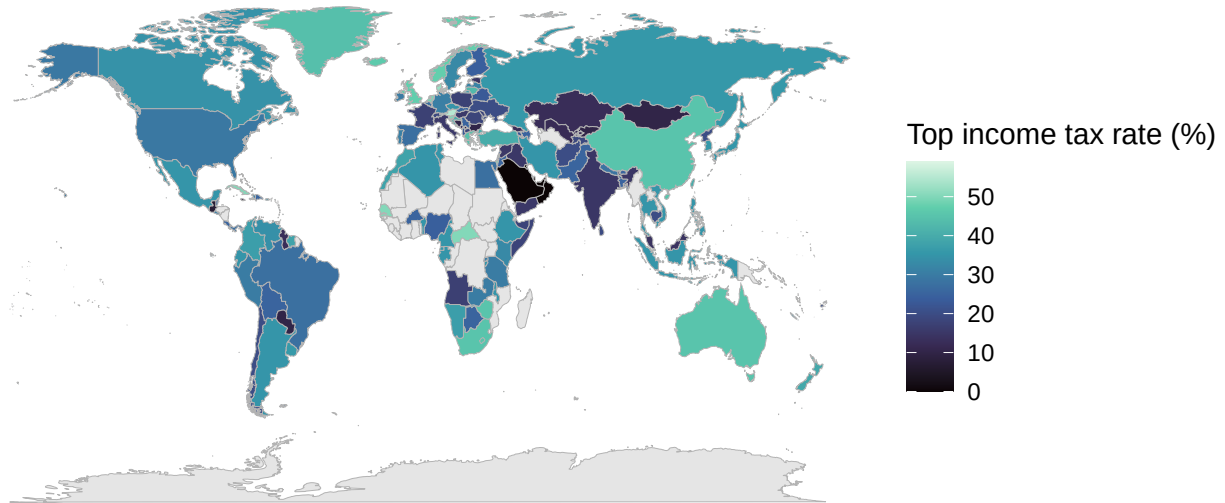
  scale_fill_viridis_c(
    option = "mako",
    na.value = "grey90",
    name = "Top income tax rate (%)"
  ) +

  labs(title = "Top Marginal Income Tax Rate by Country") +
  coord_fixed(1.3) +
  theme_void()

```

p8

Top Marginal Income Tax Rate by Country



Q9. Replace missing top income tax rates with corporate tax rates

```
# Number of missing top rates before replacement
sum(is.na(tax_with_continent$`Top Marginal Income Tax Rate`))
```

```
#> [1] 51
```

```
tax_updated <- tax_with_continent %>%
  mutate(
    `Top Marginal Income Tax Rate` = if_else(
      is.na(`Top Marginal Income Tax Rate`) & !is.na(`Corporate Tax Rate`),
      `Corporate Tax Rate`,
      `Top Marginal Income Tax Rate`
    )
  )
```

```
# Check that the number of missing values drops to zero
sum(is.na(tax_updated$`Top Marginal Income Tax Rate`))
```

```
#> [1] 0
```

Missing values in the top marginal income tax rate are imputed using the corporate tax rate whenever the latter is available. The updated variable contains no missing observations after imputation, as verified by a post-check of remaining NA values.

Q10. Visualize the updated variable

```
p10 <- ggplot(tax_updated,
  aes(x = `Top Marginal Income Tax Rate`)) +
  geom_histogram(
```

```

aes(fill = as.factor(after_stat(x))),
binwidth = 5,
boundary = 0,
color = "grey",
show.legend = FALSE
) +

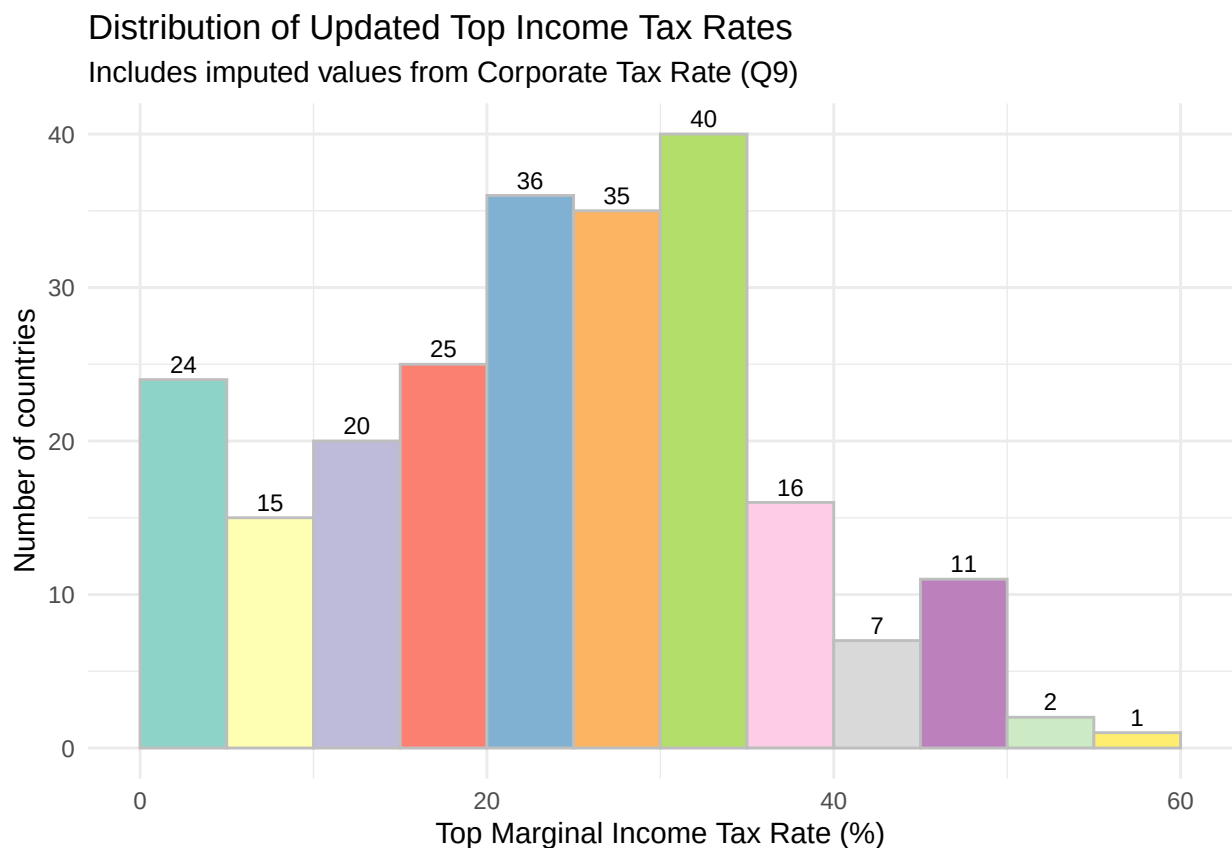
scale_fill_brewer(palette = "Set3") +

stat_bin(
  binwidth = 5,
  boundary = 0,
  geom = "text",
  aes(label = ..count..),
  vjust = -0.4,
  size = 3
) +

labs(
  title = "Distribution of Updated Top Income Tax Rates",
  subtitle = "Includes imputed values from Corporate Tax Rate (Q9)",
  x = "Top Marginal Income Tax Rate (%)",
  y = "Number of countries"
) +
theme_minimal()

```

p10



Q11.Enhanced Map: Updated Top Income Tax Rates

```

library(viridis)

map_data_updated <- world_map %>%
  left_join(

```

```

    tax_updated,
    by = c("region" = "Country")
  )
head(map_data_updated)

```

```

#>      long      lat group order region subregion Corporate Tax Rate
#> 1 -69.89912 12.45200     1     1  Aruba      <NA>                25
#> 2 -69.89571 12.42300     1     2  Aruba      <NA>                25
#> 3 -69.94219 12.43853     1     3  Aruba      <NA>                25
#> 4 -70.00415 12.50049     1     4  Aruba      <NA>                25
#> 5 -70.06612 12.54697     1     5  Aruba      <NA>                25
#> 6 -70.05088 12.59707     1     6  Aruba      <NA>                25
#> Lowest Income Tax Rate Top Marginal Income Tax Rate Capital Gains Tax Rate
#> 1                      7                      58.95                NA
#> 2                      7                      58.95                NA
#> 3                      7                      58.95                NA
#> 4                      7                      58.95                NA
#> 5                      7                      58.95                NA
#> 6                      7                      58.95                NA
#> Wealth Tax Property Tax Inheritance or Estate Tax VAT Rate Further Reading
#> 1      NA           NA                      NA      1.5 Taxation in Aruba
#> 2      NA           NA                      NA      1.5 Taxation in Aruba
#> 3      NA           NA                      NA      1.5 Taxation in Aruba
#> 4      NA           NA                      NA      1.5 Taxation in Aruba
#> 5      NA           NA                      NA      1.5 Taxation in Aruba
#> 6      NA           NA                      NA      1.5 Taxation in Aruba
#> Continent
#> 1 Americas
#> 2 Americas
#> 3 Americas
#> 4 Americas
#> 5 Americas
#> 6 Americas

```

```

P11 <- ggplot(map_data_updated) +
  geom_polygon(aes( x = long,
                    y = lat,
                    group = group,
                    fill = `Top Marginal Income Tax Rate` ),
    color = "white",
    linewidth = 0.1) +

  scale_fill_viridis_c(
    option = "B",
    na.value = "grey90",
    name = "Top marginal income tax (%)"
  ) +

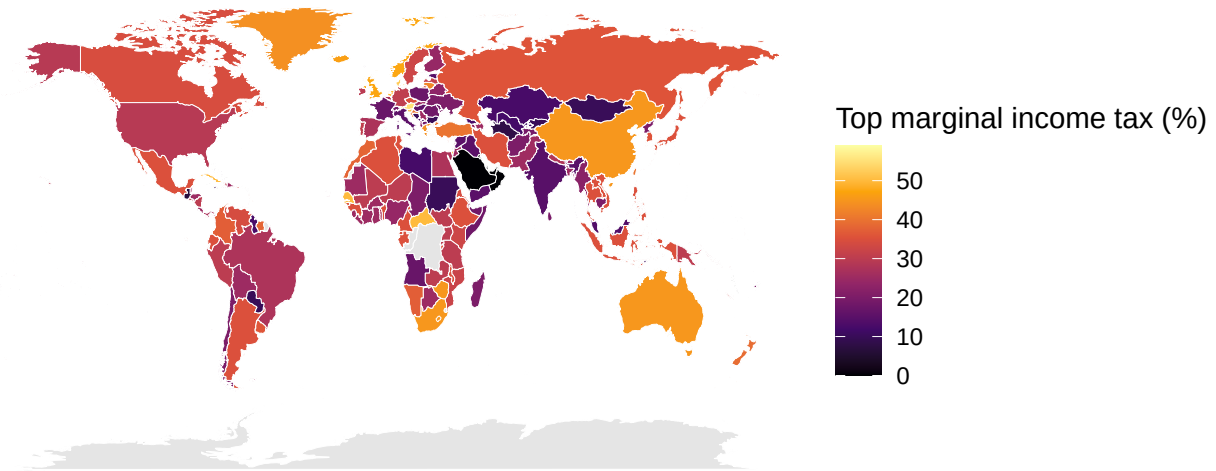
  labs(title = "Enhanced Map: Updated Top Income Tax Rates",
    fill = "Tax rate (%)") +

  coord_fixed(1.3) +
  theme_void()

```

P11

Enhanced Map: Updated Top Income Tax Rates



Part 3 – Gender wage inequality in France

We use the INSEE dataset – “Salaires dans le secteur privé selon le sexe et la catégorie socioprofessionnelle au niveau communal”, containing net monthly full-time equivalent wages in the French private sector by gender and socio-professional category (PCS) for 2022–2023.

```
library(dplyr)
library(tidyverse)
library(ggplot2)
library(patchwork)

# Load Data
df <- read.csv(
  "D:/R/2025 M2 S9 Visulisation TD/visulisation/Assign3/assign3/data part3/DS_BTS_SAL_EQTP_SEX_PCS_2023_data.csv",
  sep = ";",
  dec = ",",
  stringsAsFactors = FALSE
)

head(df)
```

```
#>      GEO GEO_OBJECT FREQ SEX PCS_ESE DERA_MEASURE
#> 1 26362 BV2022 A F _T SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 2 26324 BV2022 A _T 4 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 3 26307 BV2022 A _T 6 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 4 26362 BV2022 A _T _T SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 5 27170 BV2022 A F 4 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 6 27112 BV2022 A F 4 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> CONF_STATUS TIME_PERIOD OBS_VALUE
#> 1 F 2022 2157.284701
#> 2 F 2022 3112.937532
#> 3 F 2023 2013.097449
#> 4 F 2023 2483.037253
#> 5 F 2022 2107.221064
#> 6 F 2023 2303.18054
```

```
# 1. Data Cleaning
df_clean_final <- df %>%
  filter(PCS_ESE %in% c("6", "5", "4", "1T3"),
         SEX %in% c("F", "M")) %>%
  mutate(
    PCS_Label = factor(case_when(
      PCS_ESE == "6" ~ "Blue-collar",
      PCS_ESE == "5" ~ "White-collar",
      PCS_ESE == "4" ~ "Intermediate Prof.",
      PCS_ESE == "1T3" ~ "Executives"
    ), levels = c("Blue-collar", "White-collar", "Intermediate Prof.", "Executives")),

    #recode gender
    SEX_Label = ifelse(SEX == "F", "Female", "Male"),
    #time, wage => numeric/factor
    TIME_PERIOD = as.factor(TIME_PERIOD),
    WAGE = as.numeric(OBS_VALUE)
  )
head(df_clean_final)
```

```
#>      GEO GEO_OBJECT FREQ SEX PCS_ESE      DERA_MEASURE
#> 1 27170    BV2022    A  F      4 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 2 27112    BV2022    A  F      4 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 3 27112    BV2022    A  M      5 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 4 26307    BV2022    A  M      6 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 5 26325    BV2022    A  M     1T3 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#> 6 26362    BV2022    A  M      5 SALAIRE_NET_EQTP_MENSUEL_MOYENNE
#>  CONF_STATUS TIME_PERIOD  OBS_VALUE  PCS_Label SEX_Label  WAGE
#> 1          F      2022 2107.221064 Intermediate Prof.  Female 2107.221
#> 2          F      2023 2303.18054 Intermediate Prof.  Female 2303.181
#> 3          F      2023 1844.056156   White-collar   Male 1844.056
#> 4          F      2023 2079.049927   Blue-collar   Male 2079.050
#> 5          F      2023 4373.175294   Executives   Male 4373.175
#> 6          F      2022 1922.707102   White-collar   Male 1922.707
```

2. Efficient Data Aggregation

```
## Mean wage by occupation, gender, year
df_summary_base <- df_clean_final %>%
  group_by(PCS_Label, SEX_Label, TIME_PERIOD) %>%
  summarise(Mean_Wage = mean(WAGE, na.rm = TRUE), .groups = "drop")
df_summary_base
```

```
#> # A tibble: 16 x 4
#>   PCS_Label  SEX_Label TIME_PERIOD Mean_Wage
#>   <fct>      <chr>      <fct>      <dbl>
#> 1 Blue-collar Female    2022    1709.
#> 2 Blue-collar Female    2023    1795.
#> 3 Blue-collar Male      2022    1968.
#> 4 Blue-collar Male      2023    2057.
#> 5 White-collar Female    2022    1789.
#> 6 White-collar Female    2023    1871.
#> 7 White-collar Male      2022    1883.
#> 8 White-collar Male      2023    1962.
#> 9 Intermediate Prof. Female    2022    2266.
#> 10 Intermediate Prof. Female    2023    2347.
#> 11 Intermediate Prof. Male      2022    2584.
#> 12 Intermediate Prof. Male      2023    2676.
#> 13 Executives   Female    2022    3373.
#> 14 Executives   Female    2023    3427.
#> 15 Executives   Male      2022    3959.
#> 16 Executives   Male      2023    3974.
```

```
## Female / male wage ratio (%)
df_ratio_analysis <- df_summary_base %>%
  pivot_wider(names_from = SEX_Label, values_from = Mean_Wage) %>%
  mutate(
    FM_Ratio = (Female / Male) * 100,
    Ratio_Label = paste0(round(FM_Ratio, 1), "%")
  )
df_ratio_analysis
```

```
#> # A tibble: 8 x 6
#>   PCS_Label      TIME_PERIOD Female   Male FM_Ratio Ratio_Label
#>   <fct>         <fct>         <dbl> <dbl>   <dbl> <chr>
#> 1 Blue-collar  2022           1709. 1968.    86.8 86.8%
#> 2 Blue-collar  2023           1795. 2057.    87.2 87.2%
#> 3 White-collar 2022           1789. 1883.    95.0 95%
#> 4 White-collar 2023           1871. 1962.    95.3 95.3%
#> 5 Intermediate Prof. 2022           2266. 2584.    87.7 87.7%
#> 6 Intermediate Prof. 2023           2347. 2676.    87.7 87.7%
#> 7 Executives    2022           3373. 3959.    85.2 85.2%
#> 8 Executives    2023           3427. 3974.    86.2 86.2%
```

3. Visualization Part 1: Absolute Wage Trends

```
df_summary_base <- df_summary_base %>%
  mutate(YearGender = paste0(TIME_PERIOD, "_", SEX_Label))
df_summary_base
```

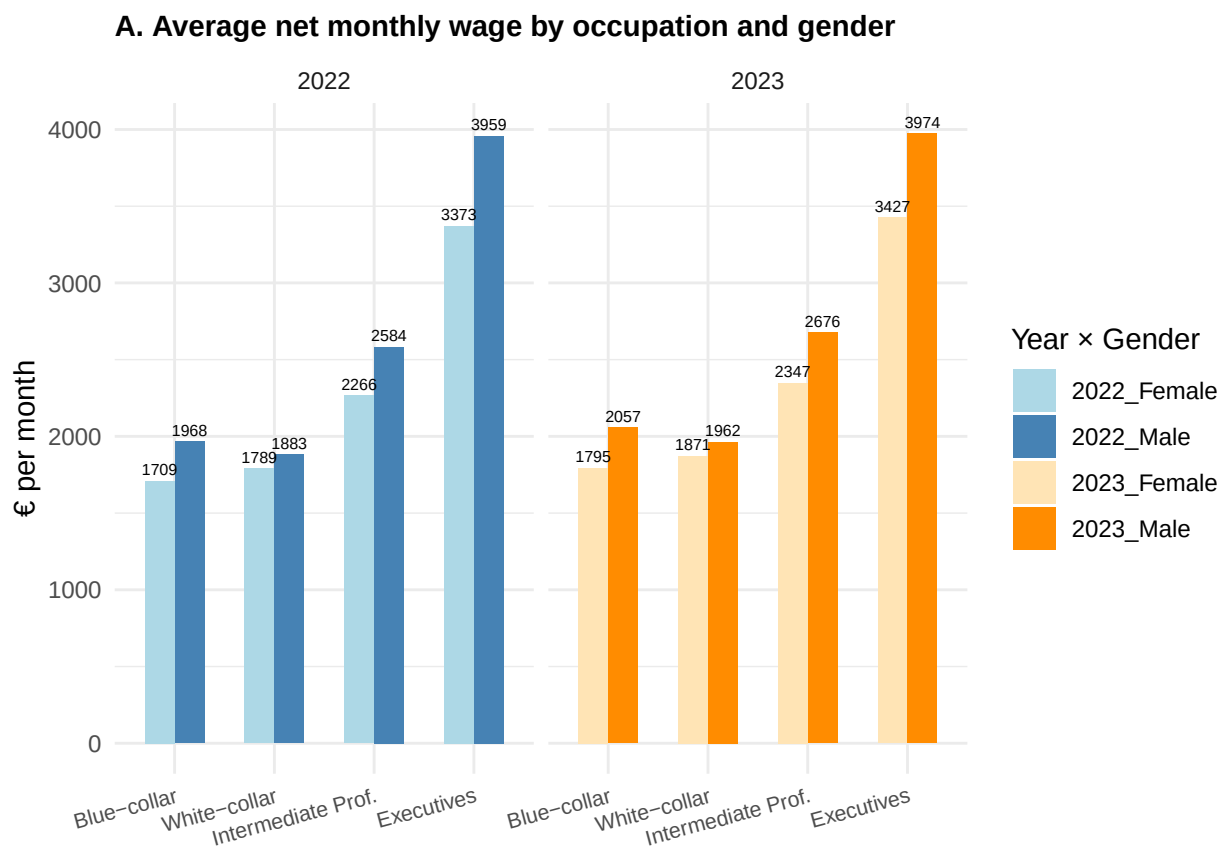
```
#> # A tibble: 16 x 5
#>   PCS_Label      SEX_Label TIME_PERIOD Mean_Wage YearGender
#>   <fct>         <chr>         <fct>         <dbl> <chr>
#> 1 Blue-collar  Female      2022           1709. 2022_Female
#> 2 Blue-collar  Female      2023           1795. 2023_Female
#> 3 Blue-collar  Male        2022           1968. 2022_Male
#> 4 Blue-collar  Male        2023           2057. 2023_Male
#> 5 White-collar Female      2022           1789. 2022_Female
#> 6 White-collar Female      2023           1871. 2023_Female
#> 7 White-collar Male        2022           1883. 2022_Male
#> 8 White-collar Male        2023           1962. 2023_Male
#> 9 Intermediate Prof. Female      2022           2266. 2022_Female
#> 10 Intermediate Prof. Female      2023           2347. 2023_Female
#> 11 Intermediate Prof. Male        2022           2584. 2022_Male
#> 12 Intermediate Prof. Male        2023           2676. 2023_Male
#> 13 Executives  Female      2022           3373. 2022_Female
#> 14 Executives  Female      2023           3427. 2023_Female
#> 15 Executives  Male        2022           3959. 2022_Male
#> 16 Executives  Male        2023           3974. 2023_Male
```

```
p1 <- ggplot(df_summary_base,
  aes(x = PCS_Label,
    y = Mean_Wage,
    fill = YearGender)) +
  geom_col(position = position_dodge(width = 0.6),
    width = 0.6) +
  facet_wrap(~ TIME_PERIOD) +

  scale_fill_manual(
    name = "Year × Gender",
    values = c(
      "2022_Female" = "lightblue",
      "2022_Male" = "steelblue",
      "2023_Female" = "moccasin",
      "2023_Male" = "darkorange"
    )
  ) +
```

```
labs(
  title = "A. Average net monthly wage by occupation and gender",
  x = NULL, y = "€ per month"
) +
geom_text(
  aes(label = round(Mean_Wage, 0)),
  position = position_dodge(width = 0.6),
  vjust = -0.5,
  size = 2
) +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 15, hjust = 1, size = 8),
  plot.title = element_text(size = 11, face = "bold")
)
```

p1



4. Visualization Part 2: Relative Ratios

```
p2_ratio <- ggplot(df_ratio_analysis, aes(x = PCS_Label,
  y = FM_Ratio,
  color = TIME_PERIOD,
  group = TIME_PERIOD)) +

  geom_hline(yintercept = 100,
    linetype = "dashed",
    color = "gray",
    linewidth = 0.8) +

  geom_line(linewidth = 1.2,
    alpha = 0.8) +

  geom_point(size = 3) +

  geom_text(aes(label = Ratio_Label),
    vjust = -1.2,
```



```

    fontface = "bold",
    size = 3,
    show.legend = FALSE) +

facet_wrap(~TIME_PERIOD) +

scale_color_manual(
  name = "Year",
  values = c(
    "2022" = "steelblue",
    "2023" = "darkorange"
  )
) +

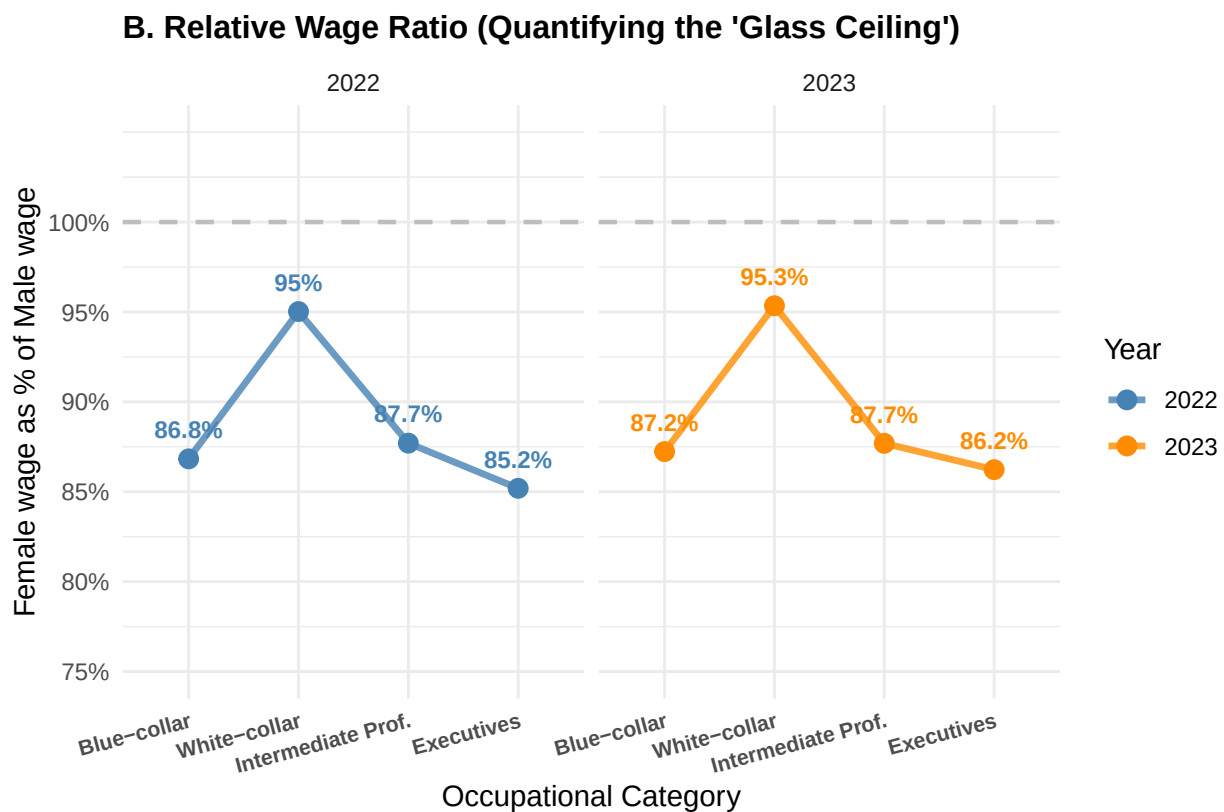
scale_y_continuous(limits = c(75, 105),
  breaks = seq(75, 100, 5),
  labels = function(x) paste0(x, "%")) +

theme_minimal() +

theme(axis.text.x = element_text(angle = 15, hjust = 1, size = 8, face = "bold"),
  plot.title = element_text(size = 12, face = "bold")) +

labs(title = "B. Relative Wage Ratio (Quantifying the 'Glass Ceiling')",
  y = "Female wage as % of Male wage", x = "Occupational Category",
  caption = "Source: Calculated based on INSEE data.")
p2_ratio

```



5. Final Combination

```

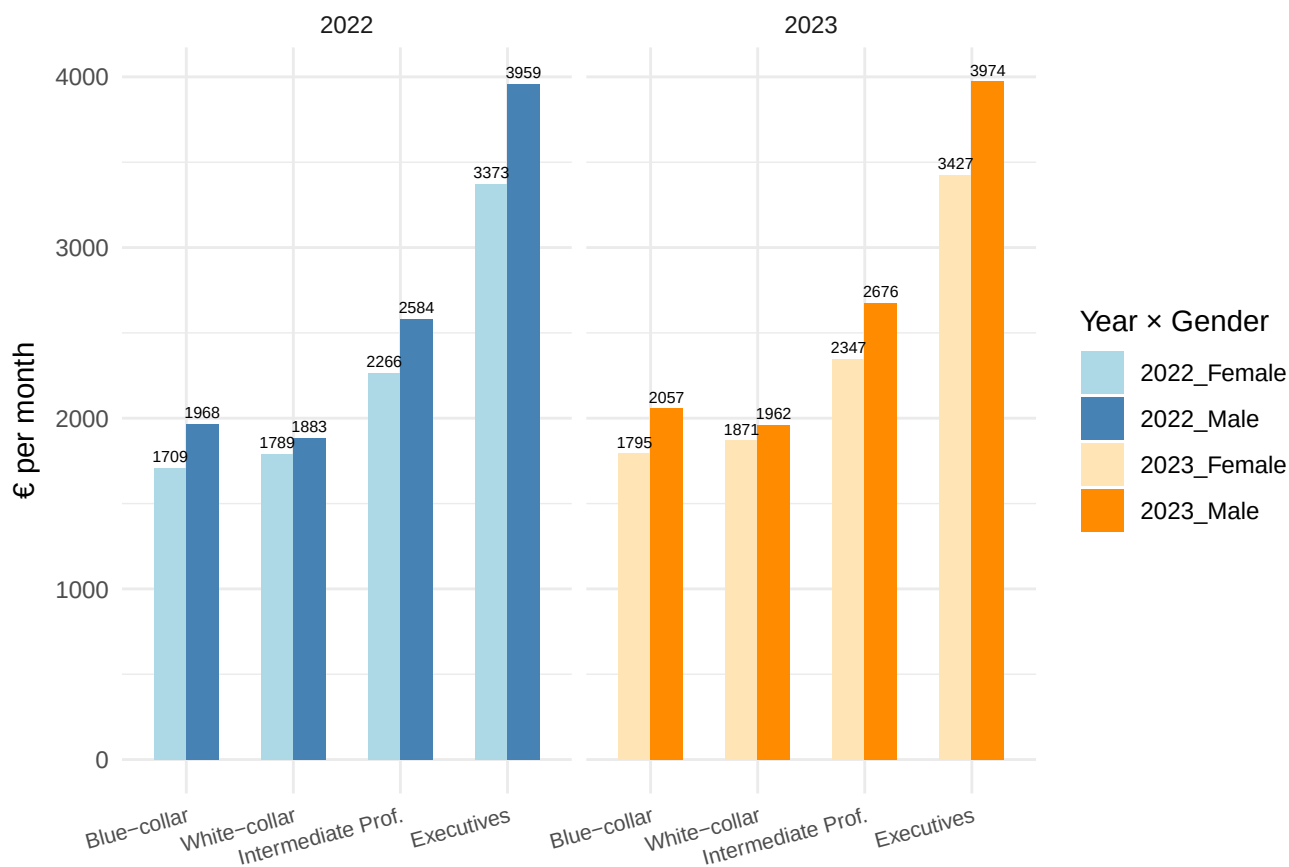
final_combined_plot <- p1 / p2_ratio +
  plot_layout(heights = c(2.5, 2)) +
  plot_annotation(
    title = "Structural Persistence of the Gender Wage Gap in France (2022-2023)",
    subtitle = "Analysis of absolute divergence and relative structural stagnation amidst inflation.",
    theme = theme(plot.title = element_text(size = 14, face = "bold", hjust = 0.5),
      plot.subtitle = element_text(size = 10, hjust = 0.5))

```

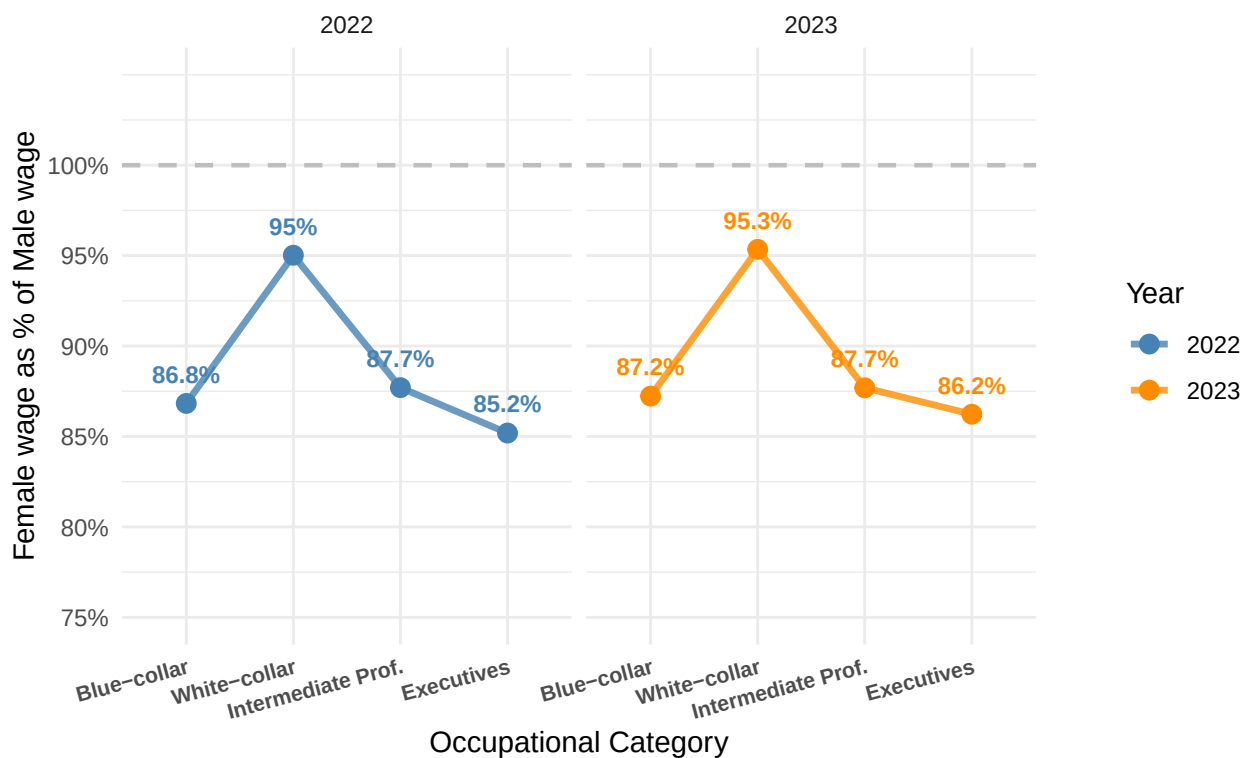
Structural Persistence of the Gender Wage Gap in France (2022–2023)

Analysis of absolute divergence and relative structural stagnation amidst inflation.

A. Average net monthly wage by occupation and gender



B. Relative Wage Ratio (Quantifying the 'Glass Ceiling')



Source: Calculated based on INSEE data.

The visualizations reveal a clear and consistent occupational structure in gender wage inequality in France.

Panel A shows that men earn more than women across all socio-professional categories, and the absolute wage gap widens as occupational status increases. The largest euro-level disparity appears among executives, where both wages and inequality are highest.

Panel B translates these differences into relative terms. Women earn close to 95% of men’s wages in white-collar jobs, but only about 85–86% among executives. Notably, these ratios remain remarkably stable between 2022 and 2023 despite overall wage growth and inflation, suggesting that the gap is structural rather than cyclical.

Taken together, the two panels point to a persistent *glass-ceiling effect*: the higher the occupational level, the wider the gender gap—both in absolute euros and as a percentage of men’s earnings. This pattern highlights long-standing structural barriers that continue to limit women’s access to higher-paid positions.